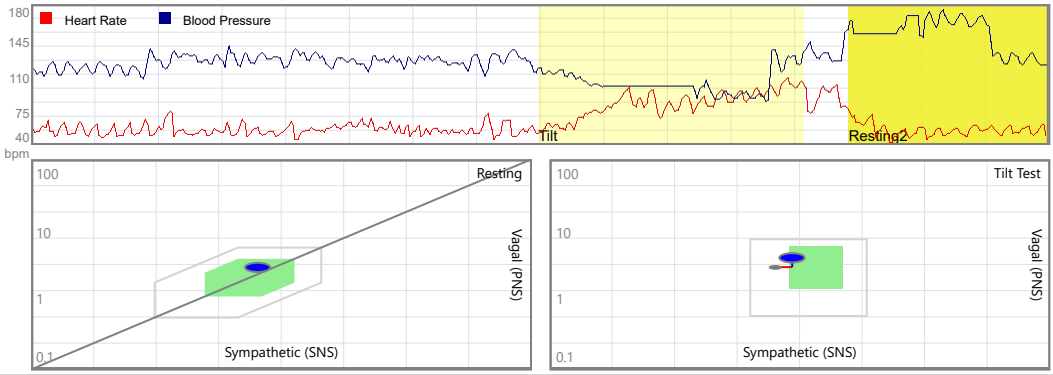
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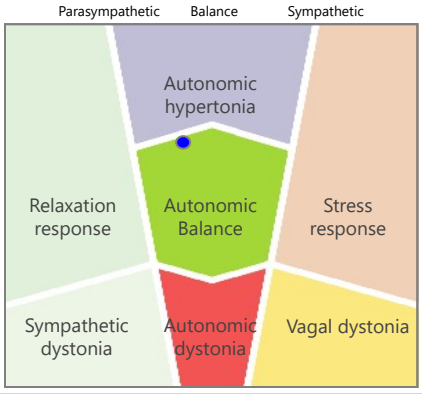
SUMMARY REPORT

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

Physician Only Report
Exam Date: Apr 11 2025 16:52

Autonomic Nervous System Function





	HR	BP	SNS	PNS	SpO2	Comment
Resting	56	124/54	2.28	2.89	97	Borderline normal level of sympathetic and parasympathetic tone;
Tilt Test	81	130/50	2.86	3.83	98	Suggests Orthostatic Intolerance; Isolated cardiovascular intolerance. This may be a sign of accelerated aging, physical exhaustion or other conditions causing a deficit in the cardiovascular tolerance to physical exertions;
Resting2	57	130/50	2.09	3.41	96	

	Score	Units	Norms		Score	Units	Norms		Score	Units	Norms
Autonomic Activity	80	ms²	55-90	Autonomic Balance	-1.2	ratio	-3 - 3	K30/15 Ratio	1.12	ratio	>1.1
Functional Age	13	years	< 19								

	Score	Norms	Description	Comments
MSI	0.39	< 1.0	MSI is an indicator of accumulative mental stress. Mental stress can be caused by situations one has to cope with daily and the emotional states that result. The way in which a person copes with stressful events is a significant contributing factor to this process. When the stress becomes too great, the body will show	The score is average. It indicates low mental stress. Stress in the recent past has had little effect on your mental-health. The negative effects of mental stress are limited.
PSI	-0.86	< 1.0	PSI is an indicator of accumulative chronic physical stress. Physical stress can be caused by long-term exposure to negative factors in one`s life or physical illness. The way in which a person copes with physical stressors is a significant contributing factor to this process. When the stress becomes too great, the body	The score is below average. It indicates low physical stress. Stress in the recent past has had little or no long-term effect on your physical-health. The negative effects of physical stress are limited.

Cardiovascular Function

	Score	Norms	Description	Comments
EEI	0.6	0.3 - 0.7	EEI is an indicator for left ventricle ejection power and elasticity of large arteries.	Borderline Normal Blood Circulation
DDI	0.58	0.3 - 0.7	DDI indicates the contractility, tension and stiffness in the small arteries.	Normal Blood Circulation
DEI	0.74	0.3 - 0.7	DEI represents the reflection of arterial elasticity and blood flow in the venous system.	Dilation of arterioles - Possible withdrawal of Sympathetic tone/Arterial vasodilator drugs.
AI	-0.24	< -0.7	Augmentation Index (AI) is a useful marker for cardiac risk. AI increases with age and a sedentary lifestyle	AI is a measure of arterial stiffness and it provides general information about the arteries. AI is positively correlated with pulsewave velocity, blood pressure and hypertension

	Score	Units	Norms		Score	Units	Norms		Score	Units	Norms
Reflection Index	0.6		.65-.85	Stiffness Index	6.33	m/s	< 8.0	Stroke Volume	140	ml	55-100
Cardiac Output	7.7	l/min	4.0-8.0	Mean Arterial Pressure	77	mmHg	70-110	Blood Volume	6.03	l	3-5
C1	12.4	ml/mmHg	> 10.0	C2	6.22	ml/mmHg	> 6.0	DPTI/SPTI	0.41	ratio	
Ventricular Extrasystole	0		< 1	Atrial Extrasystole	0		< 1	Artifacts	2		< 1
QRS	56	ms	60-120	QTc	433	ms	350-460	ST seg	234	ms	80-120
PR int	56	ms	120-200	Body Mass Index	23.2		19-25				

Physician`s Notes:

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AUTONOMIC NERVOUS SYSTEM FUNCTION

The Autonomic Nervous System (ANS) is the involuntary division of the nervous system. It consists of autonomic neurons that conduct impulses from the central nervous system (brain and/or spinal cord) to glands, smooth muscle and cardiac muscle. ANS neurons are responsible for regulating the secretions of certain glands (i.e., salivary glands) and the regulation of heart rate and peristalsis (contraction of smooth muscle in the digestive tract), among other functions. ANS regulates physiologic processes, such as blood pressure, heart rate, body temperature, digestion, metabolism, fluid and electrolyte balance, sweating, urination, defecation, sexual response, and other processes. Regulation occurs without conscious control, i.e., autonomously. Disorders of the ANS can affect any system of the body. They can originate in the peripheral or central nervous system and may be primary or secondary to other disorders. Symptoms suggesting autonomic dysfunction include orthostatic hypotension, heat intolerance, and nausea.

The ANS has two major divisions: the sympathetic and parasympathetic systems. Many organs are controlled primarily by either the sympathetic or parasympathetic system, although they may receive input from both; occasionally, functions are reciprocal.

Heart rate variability (HRV) is a measure of variations in the heart rate. It is calculated by analyzing the time series of beat-to-beat intervals from ECG. HRV is a noninvasive electrocardiographic marker reflecting the activity of the sympathetic and vagal components of the ANS on the sinus node of the heart. It expresses the total amount of variations of both instantaneous HR and RR intervals.
HRV frequency domain Fourier transform analysis, to the beat-to-beat RR interval time series. Low Frequency band (LF) derives from sympathetic activity and has been hypothesized to reflect the delay in the baroreceptor loop. High Frequency band (HF) is driven by respiration and appears to derive mainly from vagal activity or the parasympathetic nervous system.

Sympathetic Activity (SNS): 2.28

The sympathetic nervous system activates the "fight-or-flight" responses. It activates the glands and organs that defend the body against attack. Thus, sympathetic output increases heart rate and contractility, bronchodilation, hepatic glycogenolysis and glucose release, BMR (basal metabolism rate), and muscular strength. The sympathetic nervous system uses up energy and depletes the body.

Parasympathetic Vagal Activity (PNS): 2.89

The parasympathetic nervous system controls the "rest and digest" responses. Gastrointestinal secretions and motility (including evacuation) are stimulated, heart rate is slowed, and blood pressure decreases. The parasympathetic nervous system, when activated by rest, relaxation and happy thoughts, is essential for balanced living and for all healing.

Autonomic SNS/PNS Ratio: 0.79



A balance between the two branches of ANS is essential for good health. Most illnesses and injuries cause or result from an imbalance between these two branches.

Borderline normal level of sympathetic and parasympathetic tone;

Total Autonomic Activity: 80



Active Standing / Head-up Tilt

This test complements the autonomic evaluation of active standing. It is intended to evaluate for orthostatic intolerance. The patient performs active standing or head-up tilt, with monitoring of the pulse and BP. Symptoms of lightheadedness or syncope in conjunction with changes in pulse or BP constitute a positive test. In normal subjects, a decrease in BP related to the redistribution of blood due to gravity is observed, which, in turn, elicits the activation of baroreceptors. Feedback mechanisms activate the sympathetic nervous system while decreasing parasympathetic activity in order to reestablish BP levels.

Physician`s Notes:

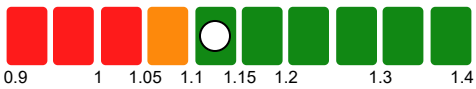
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Normal ANS response to Tilt Test;
Isolated cardiovascular intolerance. This may be a sign of accelerated aging, physical exhaustion or other conditions causing a deficit in the cardiovascular tolerance to physical exertions;

A sympathetic surge, and parasympathetic withdrawal, may lead to an increase in chronotropy, peripheral vasoconstriction, and maintenance of blood pressure.

K30/15 Ratio: 1.12



K30/15 ratio evaluate the initial cardiac responses to active standing up. This is the ratio between the shortest R wave to R wave (RR) interval around the 15th heart beat and the longest RR interval around the 30th heart beat after standing up. Atropine abolishes the bradycardiac response, indicating that the 30/15 ratio is a parameter of cardiovagal function. A fall in systolic pressure of at least 20mmHg or in diastolic pressure of at least 10mmHg within two minutes of standing or head-up tilt is considered to define orthostatic hypotension.

Suggests Orthostatic Intolerance;

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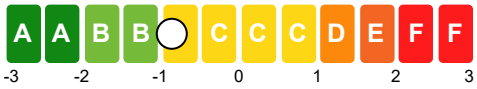
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ACCUMULATIVE STRESS ASSESSMENT

Stress is a part of everyday life. How an individual handles stress is what's important. If not handled properly, stress becomes distress. If an individual's stress is related to situations they have to cope with daily and the emotional states that result, the stress is termed mental stress. If the stress is caused by long term exposure to negative factors that result in physical illness or stress from a physical illness, this is termed accumulative physical stress.

Autonomic Physical Dysregulation (PSI): -0.86



The Physical Stress Index (PSI) is an indicator of accumulative chronic physical stress. Physical stress can be caused by long-term exposure to negative factors in one's life or physical illness. The way in which a person copes with physical stressors is a significant contributing factor to this process. When the stress becomes too great, the body will show signs of distress. Many people consider stress to be exclusively mental wherein excessive pressures, anxieties and worrying lead to feeling overburdened and under strain. However, stress includes all physical, emotional, mental, chemical and biochemical deviations from homeostasis (physiological stability). Physical stress in humans arises as a result of our body's response to physical stressors like work, noise, cold, illness, or exercise. Other lifestyle or social factors such as smoking, immobility, heavy drinking and lack of sleep produce considerable stress in the body. When the total amount of stress is greater than your body can adequately compensate for, the signs and symptoms of stress begin to appear. Signs of physical stress can include muscle tension, shallow breathing, hypertension, headaches, and a rapid heartbeat which when left untreated can result in a number of ailments including heart disease, ulcers, asthma and rheumatism.

The score is below average. It indicates low physical stress. Stress in the recent past has had little or no long-term effect on your physical-health. The negative effects of physical stress are limited.

Accumulative Mental Stress (MSI): 0.39



The Mental Stress Index (MSI) is an indicator of accumulative mental stress. When the stress becomes too great, the body will show signs of distress. Most people are abundantly familiar with mental stress as modern life tends to generate copious amounts of it. People recognize a state of having too much expected of them, of being under pressure or strain, of being barely able to cope with some external demand which is both excessive and prolonged. It has a number of synonyms, but they all carry the connotation of unreasonable demands being placed on the individual in an emotional, mental or psychological sense. In short, psychological or mental stress refers to the emotional and physiological reactions experienced when an individual confronts a situation in which the demands go beyond their coping resources.
Mental stress can manifest itself in behavioral, cognitive or emotional symptoms which may range from poor judgment to moodiness to depression. If left unchecked, mental stress can accumulate, thereby increasing its impact on overall health. The body's ability to cope decreases and the risk of further complications to both mental and physical well-being increases. Research indicates that mental stress is intimately linked to illness and disease. Constant and cumulative stress can compromise the immune system and mental well-being which increases susceptibility, decreases the strength of the body's natural defenses and affects thought processes.

The score is average. It indicates low mental stress. Stress in the recent past has had little effect on your mental-health. The negative effects of mental stress are limited.

Functional Age: 13 years

Physician`s Notes:

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VASCULAR ASSESSMENT

Ankle-Brachial Index (ABI)

ABI test is a quick, noninvasive way to check your risk of peripheral artery disease (PAD). Peripheral artery disease is a condition in which the arteries in your legs or arms are narrowed or blocked. People with peripheral artery disease are at an increased risk of heart attack, stroke, poor circulation and leg pain. ABI test compares your blood pressure measured at your ankle with your blood pressure measured at your arm. A low ankle-brachial index number can indicate narrowing or blockage of the arteries in your legs, increasing your risk of circulatory problems, and possibly causing heart disease or stroke.

*Treatment for peripheral artery disease has two major goals: Manage symptoms, such as leg pain and stop the progression of atherosclerosis throughout your body to reduce your risk of heart attack and stroke.ABI > 0.9 and 1.0 or ABI> 1.4 and TBI >= 0.5:Lifestyle changes will be helpful. If you smoke, quit to reduce your risk of complications. Walking or doing other exercise on a regular basis can improve your symptoms dramatically. Avoid certain cold medications (constrict your blood vessels and may increase your PAD symptoms).ABI < = 0.9 and > 0.6 or ABI> 1.4 and TBI < 0.5:If you have signs or symptoms of PAD, you likely will need additional medical treatment. Refer to the vascular specialist for additional exams: Ultrasound, Angiography and blood tests A sample of your blood can be used to measure your cholesterol and triglycerides and to check for diabetes.If Cholesterol lab test is abnormal: suggested cholesterol-lowering medications.If your blood pressure >140 mmHg: Suggested high blood pressure medications.If you are diabetic: Suggested medication to control blood sugar.In all cases: Suggested medications to prevent blood clots such as clopidogrel (Plavix).Symptom-relief medications. The drug cilostazol increases blood flow to the limbs both by keeping the blood thin and by widening the blood vessels. If side effect the alternative medication to cilostazol is pentoxifylline (Pentoxil).ABI < = 0.6 and TBI < 0.5:Refer to the vascular specialist for additional exams: Ultrasound, Angiography and blood tests (cholesterol, triglycerides and Glycemia/ Hb1aC). The vascular surgeon will decide if Angioplasty and surgery is necessary.*Aboyans V. et al. Measurement and Interpretation of the Ankle-Brachia I Index: A Scientific Statement from the American Heart Association. Circulation, November 2012.*

Physician`s Notes:

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ENDOTHELIAL DYSFUNCTION - PULSE WAVE VELOCITY ANALYSIS (PWV)

PWV gives you a quick and objective answer to how the vascular system is working. PWV is an excellent analysis to evaluate vascular endothelial dysfunction. This represent the elasticity of the artery. Arteries that are atherosc lerotic, arteriosclerotic, or hardened (having reduced elasticity and increased narrowing) place an extra strain on the heart, valves, and arteries which can lead to stroke, heart attack, kidney failure and/or sudden death.
The pulse wave is a physiological phenomenon, observable and measurable in the arterial system during blood circulation. During one heart systole a certain blood volume is expelled. This propagates through the arteries due to the reciprocal transformation between kinetic energy of a segment of the expelled blood volume and the potential energy of a stretched segment of the resilient vascular wall. The changes in pressure, blood flow, velocity and profile throughout the whole pulse wave. It can be used for classification of the artery elasticity.

Ejection Elasticity Index (EEI): 0.6



EEI is an indicator for left ventricle ejection power and elasticity of the large arteries. Possible causes include: hypertension, obesity, metabolic syndrome, diabetes, reduced left ventricular ejection fraction and atherosclerosis.

Borderline Normal Blood Circulation

Dicrotic Dilation Index (DDI): 0.58



DDI indicates the contractility and stiffness in the small arteries. Possible causes include: hypertension, obesity, metabolic syndrome, diabetes, and arteriosclerosis.

Normal Blood Circulation

Dicrotic Elasticity Index (DEI): 0.74



DEI represents the blood flow in the venous system and elasticity of the peripheral arteries. Possible causes may include: hypertension, obesity, metabolic syndrome, diabetes, arteriosclerosis and Raynaud's phenomenon/disease.

Dilation of arterioles - Possible withdrawal of Sympathetic tone/Arterial vasodilator drugs.

Augmentation Index (AI): -0.24

Augmentation Index measures arterial stiffness. Arteries stiffen for a number of reasons including age, high cholesterol, smoking and diabetes.
The stiffer and more hardened an artery is, the more difficult it is for blood to flow through the body. To compensate for inefficient blood flow, the heart must work harder.
Augmentation index is calculated using pulse wave velocity. A ratio is calculated from the blood pressure waveform derived from PWV analysis. AI is widely used as a measure of the augmentation of central aortic pressure by a reflected pulse wave.

Pulse Oximetry (SpO2): 97.1 %

Pulse oximetry measures the amount of oxygen in the blood through a finger clip test. A range of 96% to 100% oxygen in the blood is generally considered normal. Anything below 90% could quickly lead to life-threatening complications. Knowing this is very useful for patients with respiratory and/or cardiac problems, especially those with Chronic Obstructive Pulmonary Disease (COPD). It is also useful in diagnosing sleep disorders such as sleep apnea. Being anemic may be a cause of low oxygen saturation.

Physician`s Notes:

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ECG ASSESSMENT

The normal heart rate and rhythm are determined by the sino-atrial node in the right atrium. This node discharges electric current through the atria causing them to contract. The electric current then passes through the atrioventricular (AV) node which lies within the lower interatrial septum. Electrical impulses pass, and excite the ventricular muscles causing their contraction.

Extrasystoles (premature beats) are essentially extra beats, or contractions, which interrupt the normal regular rhythm of the heart. They occur when there is electrical discharge from somewhere in the heart other than the sino-atrial node. They are classified as atrial or ventricular extrasystoles according to their site of origin.

Extrasystoles can occur frequently in people with completely normal hearts and often do not cause any problems. In a healthy heart, extra beats occur in the following situations: excessive consumption of coffee, tobacco abuse, emotions, stress and chemical stimulants.
However, they can also be a feature of certain cardiac diseases, coronary artery disease, cardiomyopathy, mitral regurgitation may lead to arrhythmia.

Ventricular Extrasystole (PVC): 0

Atrial Extrasystole (PAC): 0

Artifacts: 2

Physician`s Notes:

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METABOLIC REPORT

BMI is a useful measure of overweight and obesity. It is calculated from your height and weight.
BMI is an estimate of body fat and a good gauge of your risk for diseases that can occur with more body fat. The higher your BMI, the higher your risk for certain diseases such as heart disease, high blood pressure, type 2 diabetes, gallstones, breathing problems, and certain cancers. Although BMI can be used as a guide, it does have some limits:
- It may overestimate body fat in athletes and others who have a muscular build.
- It may underestimate body fat in older persons and others who have lost muscle.
- It does not measure a person's metabolism Unfortunately, obesity has become an epidemic in our society. For people who are considered obese (BMI greater than or equal to 30) or those who are overweight (BMI of 25 to 29.9) and have two or more risk factors (such as hypertension, high blood sugar, inactivity, cigarette smoking), it is recommended that you lose weight. People who are overweight, do not have a high waist measurement, and have fewer than two risk factors may need to prevent further weight gain rather than lose weight.
The good news is even a small weight loss (between 5 and 10 percent of your current weight) will help lower your risk of developing those diseases. Obesity, metabolic syndrome, and diabetes are all on the rise and play a significant part as risk factors for cardiovascular disease.

Risk of Associated Disease According to BMI and Waist Size

18.5 or less: Underweight - N/A
19 - 25: Normal - very low risk of associated diseases
26 - 29: Overweight - prone to health risks
30 - 40: Overweight to Obese - high risk of associated diseases
40 or greater: Extremely Obese - very high risk of associated diseases

Body Mass Index (BMI): 23.2



Body mass index is defined as the individual's body weight divided by the square of his or her height. The body mass index can be used to identify if you are overweight.

Normal - very low risk of associated diseases

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AUTONOMIC NERVOUS SYSTEM FUNCTION

Sympathetic Activity (SNS): 2.28

Normal range: 0.8-4

Parasympathetic Vagal Activity (PNS): 2.89

Normal range: 0.8-4

Autonomic SNS/PNS Ratio: 0.79

Normal range: 0.4-2

Borderline normal level of sympathetic and parasympathetic tone;

Total Autonomic Activity: 80

Normal range: 55-90

Active Standing / Head-up Tilt

Normal ANS response to Tilt Test;
Isolated cardiovascular intolerance. This may be a sign of accelerated aging, physical exhaustion or other conditions causing a deficit in the cardiovascular tolerance to physical exertions;

K30/15 Ratio: 1.12

Normal range: >1.1

Suggests Orthostatic Intolerance;

ACCUMULATIVE STRESS ASSESSMENT

Autonomic Physical Dysregulation (PSI): -0.86

Normal range: < 1.0

The score is below average. It indicates low physical stress. Stress in the recent past has had little or no long-term effect on your physical-health. The negative effects of physical stress are limited.

Accumulative Mental Stress (MSI): 0.39

Normal range: < 1.0

The score is average. It indicates low mental stress. Stress in the recent past has had little effect on your mental-health. The negative effects of mental stress are limited.

VASCULAR ASSESSMENT

Ankle-Brachial Index (ABI)

ENDOTHELIAL DYSFUNCTION - PULSE WAVE VELOCITY ANALYSIS (PWV)

Physician`s Notes:

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Ejection Elasticity Index (EEI): 0.6

Normal range: 0.3 - 0.7

Borderline Normal Blood Circulation

Dicrotic Dilation Index (DDI): 0.58

Normal range: 0.3 - 0.7

Normal Blood Circulation

Dicrotic Elasticity Index (DEI): 0.74

Normal range: 0.3 - 0.7

Dilation of arterioles - Possible withdrawal of Sympathetic tone/Arterial vasodilator drugs.

Augmentation Index (AI): -0.24

Normal range: < -0.7

Pulse Oximetry (SpO2): 97.1 %

Normal range: > 95

ECG ASSESSMENT

Ventricular Extrasystole (PVC): 0

Normal range: < 1

Atrial Extrasystole (PAC): 0

Normal range: < 1

Artifacts: 2

Normal range: < 1

METABOLIC REPORT

Body Mass Index (BMI): 23.2

Normal range: 19-25

Normal - very low risk of associated diseases

Physician`s Notes:

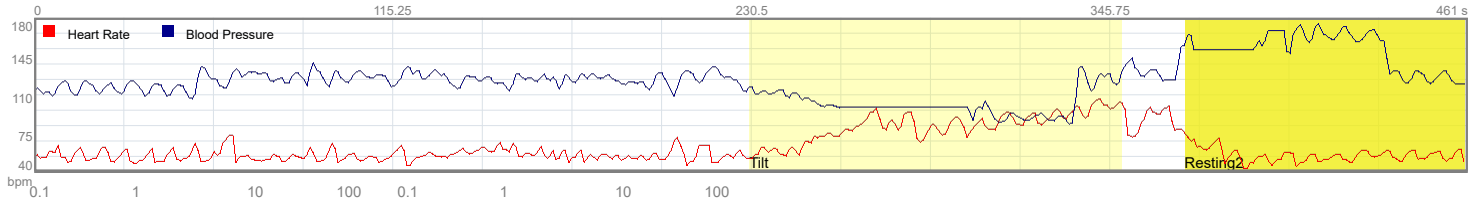
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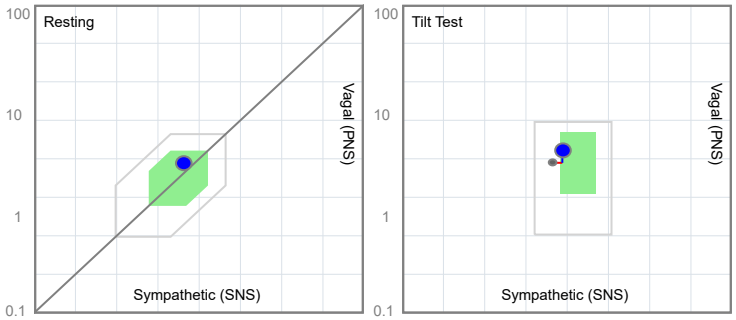
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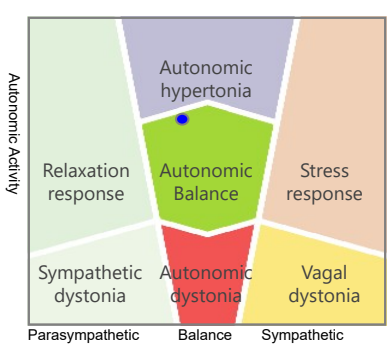
AUTONOMIC NERVOUS SYSTEM FUNCTION

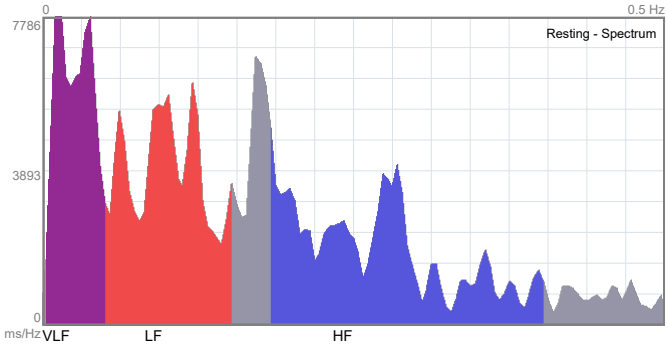
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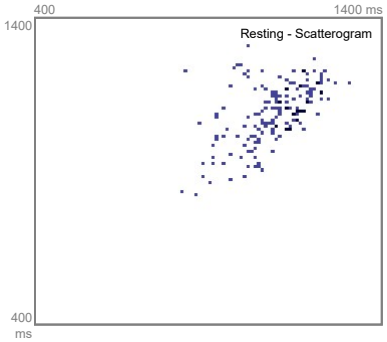
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	Duration	HR	BP	RMF	SNS	PNS	SNS/PNS	LF/HF	TSP	SDNN	rmsSD	SpO2
Resting	180	56	124/54	0.15	2.28	2.89	0.79	4.11	6612	80	61	97
Tilt Test	119	81	130/50	0.13	2.86	3.83	0.75	5.58	16056	123	38	98
Resting2	89	57	130/50	0.12	2.09	3.41	0.61	11.58	11357	127	60	96

Signal Quality: Resting: Fair Deep Breathing: Results compromised Valsalva: Results compromised Tilt Test: Good Resting2: Good

Autonomic Activity: 80.44 ∧ K30/15 Ratio: 1.12 ✓

Resting: Borderline normal level of sympathetic and parasympathetic tone;

Tilt Test: Suggests Orthostatic Intolerance; Isolated cardiovascular intolerance. This may be a sign of accelerated aging, physical exhaustion or other conditions causing a deficit in the cardiovascular tolerance to physical exertions;

Physician`s Notes:

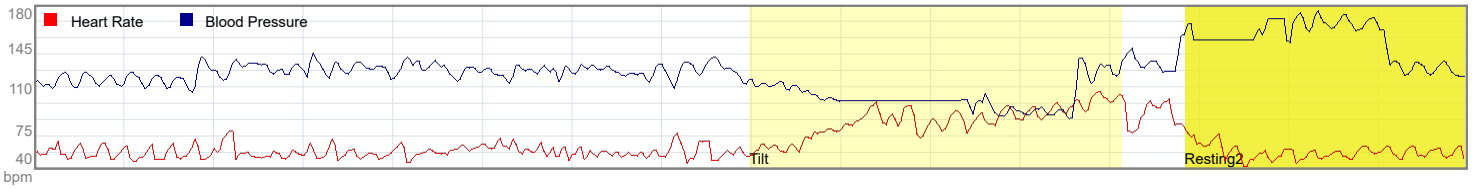
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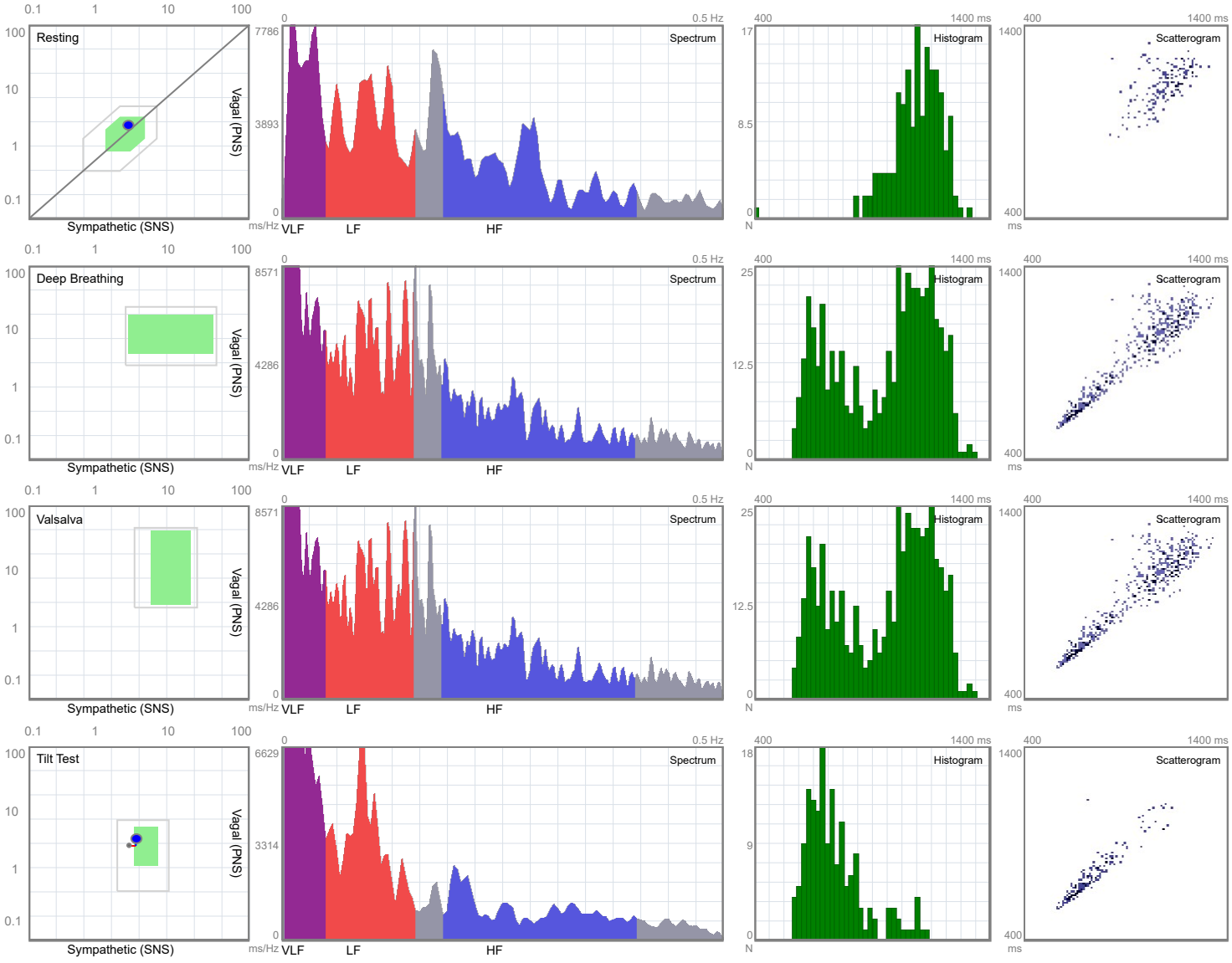
VitalScan

Autonomic Nervous System Function - Graphics

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

Physician Only Report
Exam Date: Apr 11 2025 16:52





Physician`s Notes:

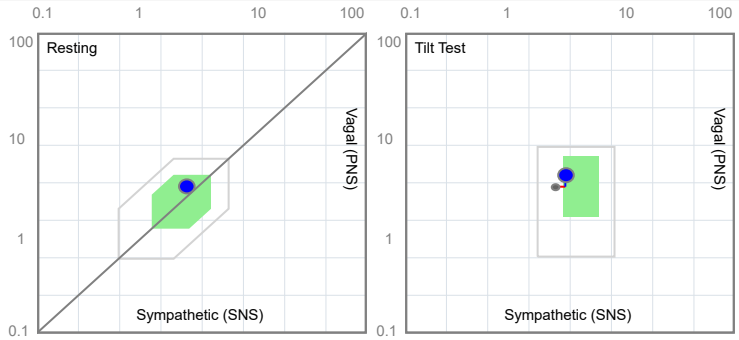
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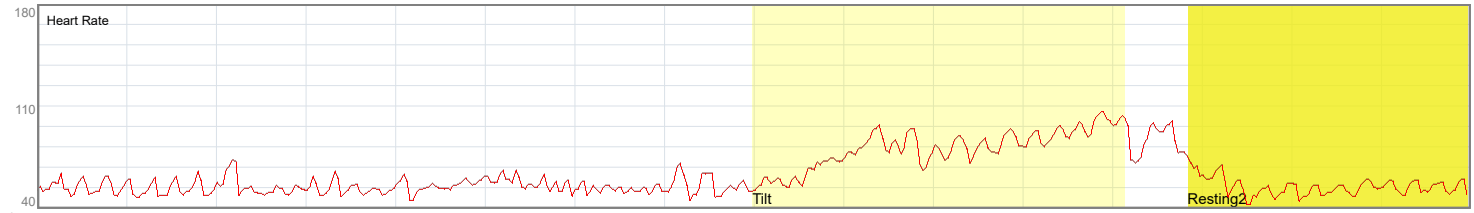
VitalScan

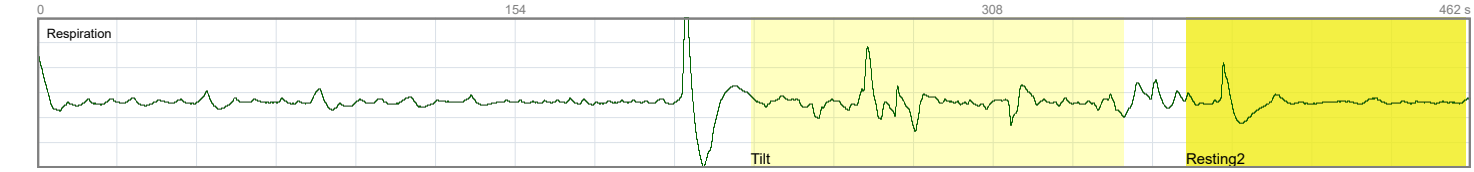
AUTONOMIC NERVOUS SYSTEM FUNCTION

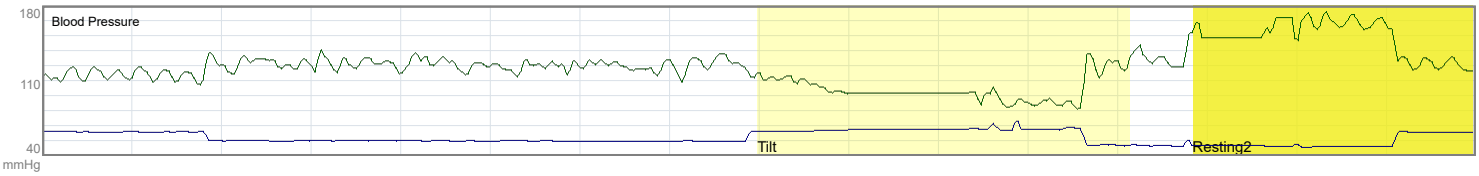
Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
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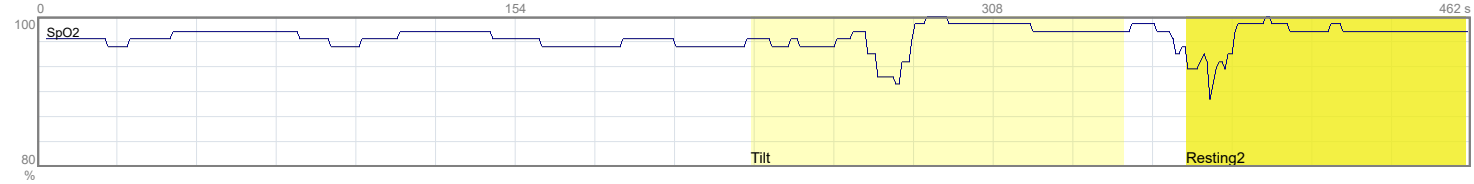
Physician Only Report
Exam Date: Apr 11 2025 16:52

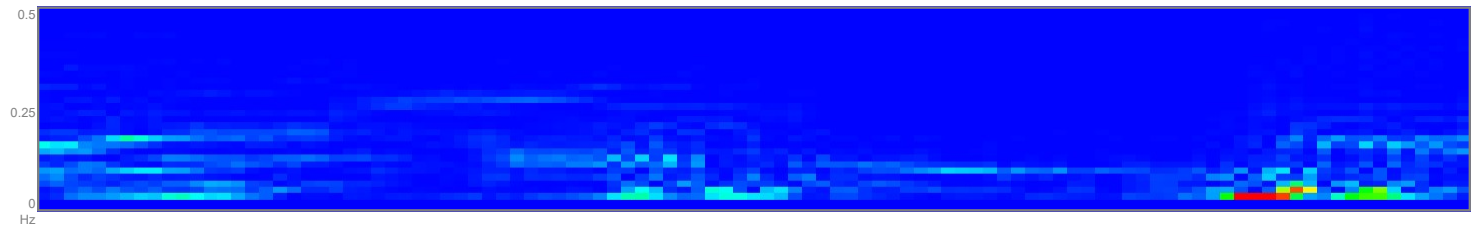


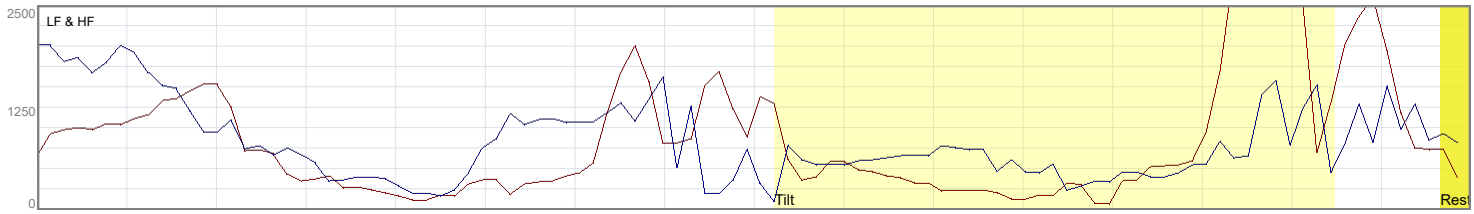












Physician`s Notes:

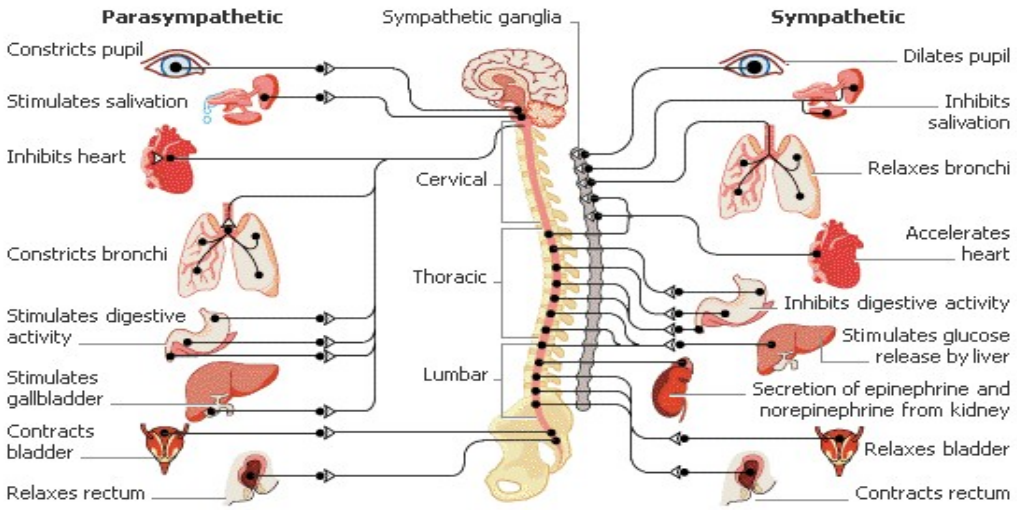
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VitalScan

AUTONOMIC NERVOUS SYSTEM FUNCTION

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
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Physician Only Report
Exam Date: Apr 11 2025 16:52



Non-Invasive Autonomic Nervous System Monitoring

Your nervous system is comprised of three parts:
1) Somatic (or sensory) nervous system
2) Motoric Nervous System
3) Autonomic Nervous System (ANS)

Your ANS is that part of your nervous system which functions to sustain your life by helping to control your heart, lungs, digestive system, blood pressure, immune system, certain reflexes, such as coughing and gagging, fluid balance, pupil diameter, sweating and sexual function.

Your ANS Consists of Two Parts (branches). There are two parts, or branches to your ANS
1) Sympathetic branch
2) Parasympathetic branch

Generally, the sympathetic branch is more in control when you are stressed, nervous, or excited, while the parasympathetic branch is more in control when you are relaxing, sleeping, or recovering from an illness or injury.
A balance between the two branches of your ANS is essential for good health. In fact, most illnesses and injuries cause or result from an imbalance between these two branches. An imbalance in your ANS can tell your doctor many things about how healthy you are as well as what can be done to keep you as healthy as possible.

ANS monitoring records your heart rate variability and respiratory activity. Your heart rate variability and respiratory activity are analyzed by a computer to determine how your ANS is controlling your heart and your lungs and other parts of your body. Your physician then interprets your results produced by the computer.

What is Heart Rate Variability (HRV)?
Heart rate variability (HRV) is a measure of your heart's ability to quickly respond to changes in your level of activity. Moderate variability is healthy. Too much or too little is unhealthy. ANS monitoring using HRV can provide your doctor with information that cannot be seen using other measures, for example on an EKG recording.

Why is including respiratory activity important?
Analyzing your respiratory activity along with your HRV is key to monitoring the balance between the two branches of your autonomic nervous system.

Why is ANS monitoring important?
Everyone is different, and so is their autonomic nervous system. How an individual responds to disease, injury, medicines and medical treatments largely depends on her or his ANS. By monitoring your ANS, your doctor can determine whether she or he is doing enough to keep you healthy. In this way, your doctor can better care for you and better maintain your well-being.

Who should be monitored?
Everyone from children to older adults can and should have their ANS monitored. Frequent monitoring, when indicated, helps your doctor better maintain your health and well-being and tailor treatments, including medications, especially for you.

Physician`s Notes:

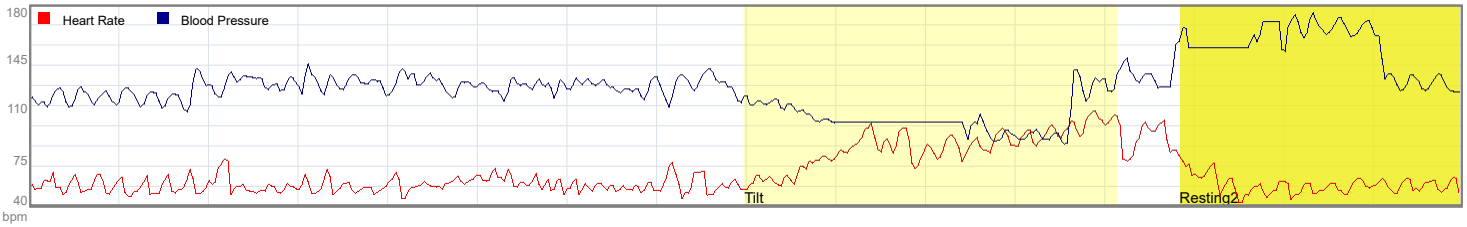
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VitalScan

AUTONOMIC NERVOUS SYSTEM FUNCTION

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

Physician Only Report
Exam Date: Apr 11 2025 16:52



	Duration	HR	BP	RMF	SNS	PNS	SNS/PNS	LF/HF	TSP	SDNN	rmsSD	SpO2
Resting	180	56	124/54	0.15	2.28	2.89	0.79	4.11	6612	80	61	97
Tilt Test	119	81	130/50	0.13	2.86	3.83	0.75	5.58	16056	123	38	98
Resting2	89	57	130/50	0.12	2.09	3.41	0.61	11.58	11357	127	60	96

Signal Quality: Resting: Fair Deep Breathing: Results compromised Valsalva: Results compromised Tilt Test: Good Resting2: Good

Autonomic Activity: 80.44 ^ K30/15 Ratio: 1.12 ✓

RESTING:
Borderline normal level of sympathetic and parasympathetic tone;

TILT TEST:
Suggests Orthostatic Intolerance; Isolated cardiovascular intolerance. This may be a sign of accelerated aging, physical exhaustion or other conditions causing a deficit in the cardiovascular tolerance to physical exertions;

Physician`s Notes:

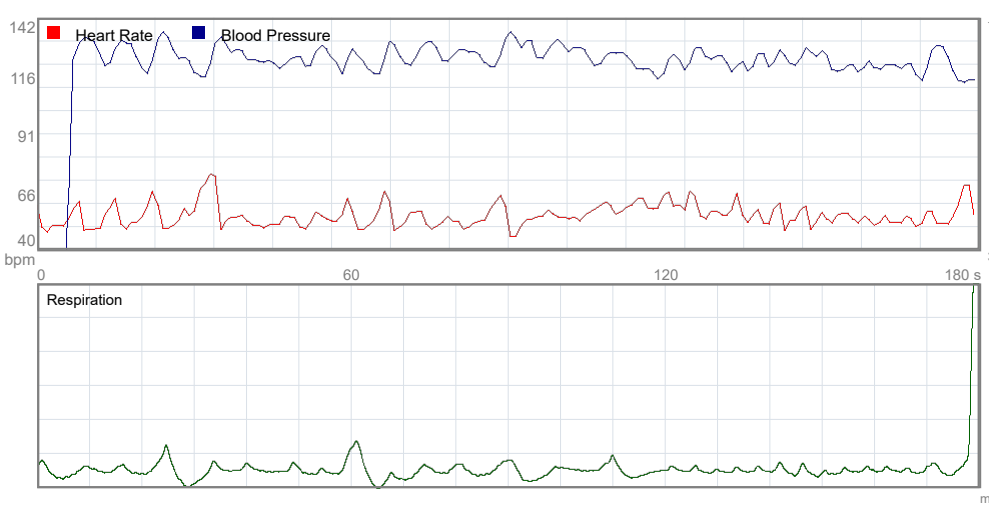
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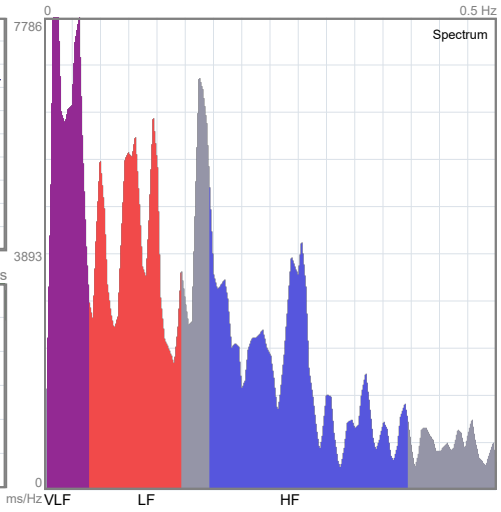
VitalScan

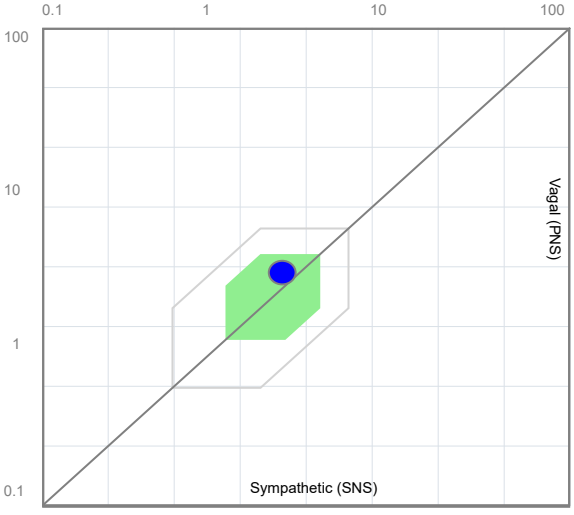
CARDIAC AUTONOMIC ASSESSMENT - RESTING

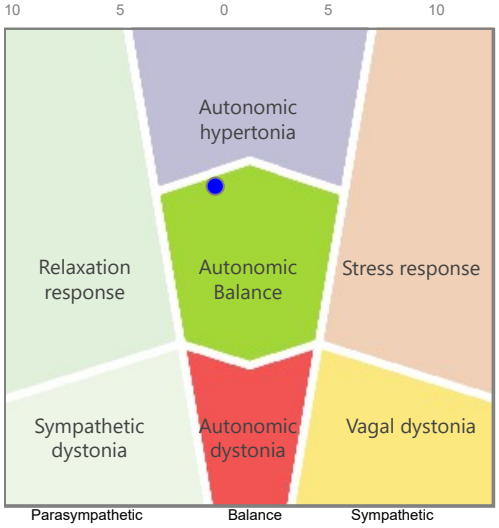
Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

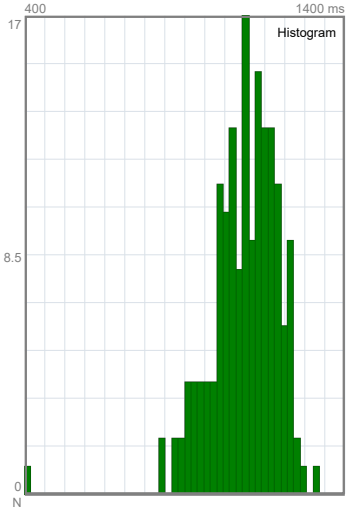
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Exam Date: Apr 11 2025 16:52











Heartbeats: 151 Artifacts: 9 (6%) Signal Quality: Results possibly compromised due to high number of artifacts!

Autonomic Activity: 80.44 ^

	value	norms	borderline	low	high
SNS	2.28	1.1-3.8	0.6-1.1	<0.6	>5.5
PNS	2.89	1.1-3.8	0.6-1.1	<0.6	>5.5
SNS/PNS	0.79	0.4-3		<0.4	>3

HR	56 bpm	**BP**	124/54 mmHg	**RMF**	0.15 Hz	**VLF**	2845 ms^2
LF	1980 ms^2	**HF**	482 ms^2	**SDNN**	80 ms	**rmsSD**	61 ms
pNN50	49 %	**TSP**	6612 ms^2	**LF/HF**	4.11	**SpO2**	97

HR,PD - bpm; RMF - Hz; VLF,LF,HF,TSP - ms^2; SNS,PNS - bpm^2

Borderline normal level of sympathetic and parasympathetic tone;

Physician`s Notes:

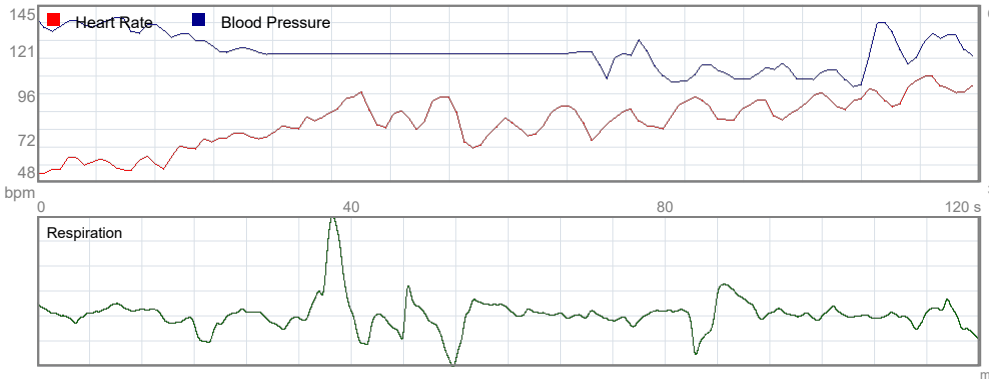
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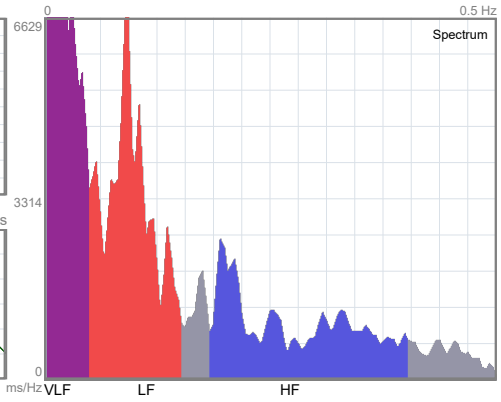
VitalScan

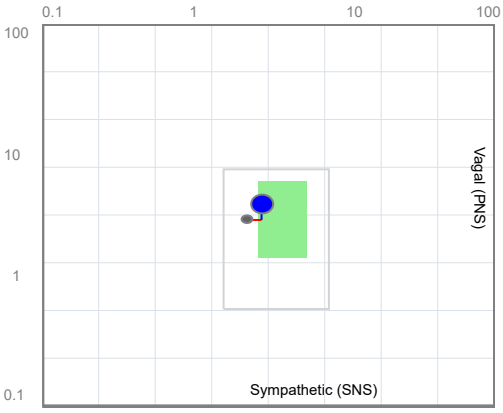
Cardiac Autonomic Assessment - Tilt Test

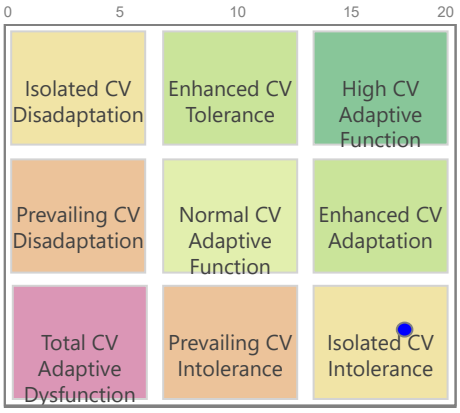
Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

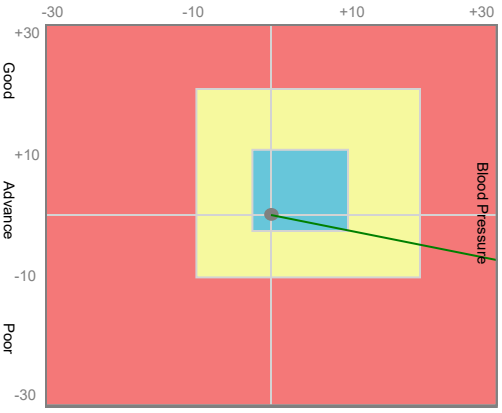
Physician Only Report
Exam Date: Apr 11 2025 16:52

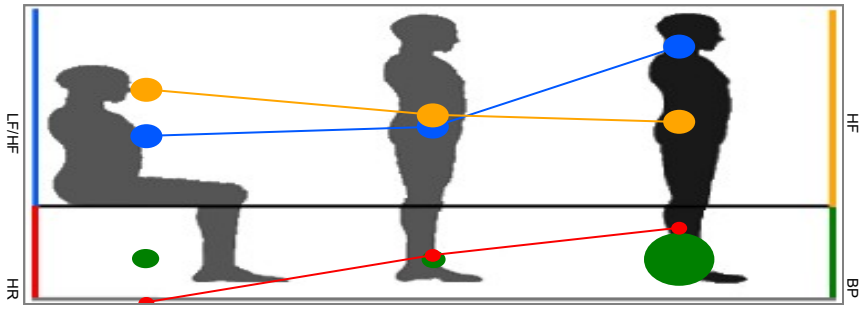












	Resting	Tilt 1m	Tilt 2m
HR	56	77	89
LF%	29.95	24.57	39.19
HF%	7.29	4.58	2.96
SDNN	80.47	116.04	66.75
BP sys	124	113	130
LF/HF	4.11	5.36	13.26

Heartbeats: 161 Artifacts: 1 (0.6%) Signal Quality: Good

	Resting	Tilt	ratio (st/re)	norms	borderline	high	low
xPNS	2.89	3.83	1.33	<2	2-2.5	>2.5	
xSNS	2.28	2.86	1.25	1.2-2.5	2.5-3.5	>3.5	<0.7
HR	56	81	1.45	>1.1	1-1.1	>30	<1
HR 30/15 Ratio		1.12 (62-55)		>1.1			<1.1
BP 30/15 Ratio		1.1 (140-127)		>1.1			<1.1

TSP	16056 ms²	**LF/HF**	5.58	**SDNN**	123 ms	**rmsSD**	38 ms
pNN50	13 %	**SDSD**	37.79 ms	**PH**	80 mmHg	**SpO2**	98 %

HR,PD - bpm; RMF - Hz; VLF,LF,HF,TSP - ms²; SNS,PNS - bpm²

Suggests Orthostatic Intolerance; Isolated cardiovascular intolerance. This may be a sign of accelerated aging, physical exhaustion or other conditions causing a deficit in the cardiovascular tolerance to physical exertions;

Physician`s Notes:

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VitalScan

ECG ANALYSIS REPORT

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

Physician Only Report
Exam Date: Apr 11 2025 16:52

Recorded time	
Start Date	2025-04-11 16:52:05
Duration	7 min 42 sec (488 beats)
High Pass Filters	3.4 Hz

Heart Rate	
Average Heart Rate	63 bpm
Fastest rate	106 bpm
Slowest rate	45 bpm

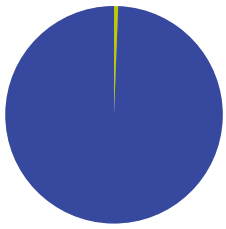
Ventricular Details	
PVC - Ventricular Ectopy	0 beats (0%)
Ventricular Couplet	0 episodes

Supraventricular Details	
PAC - Supraventricular Ectopy	0 beats (0%)
Supraventricular Couplet	0 episodes

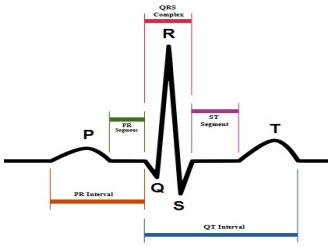
Pause / Block	
Irregular / Artifact beat	3 beats (0.6%)

HRV Analysis	
SDNN	89 ms

QRS Analysis	
QRS	55 ms
QT / QTc	430 ms / 452 ms
PR int / seg	77 ms / 43 ms
ST int / seg	401 ms / 239 ms



PAC - Atrial Extrasystole: 0%
PVC - Ventricular Extrasystole: 0%
Irregular / Artifact beat: 0.6%
Normal: 99.4%



Interpretations:



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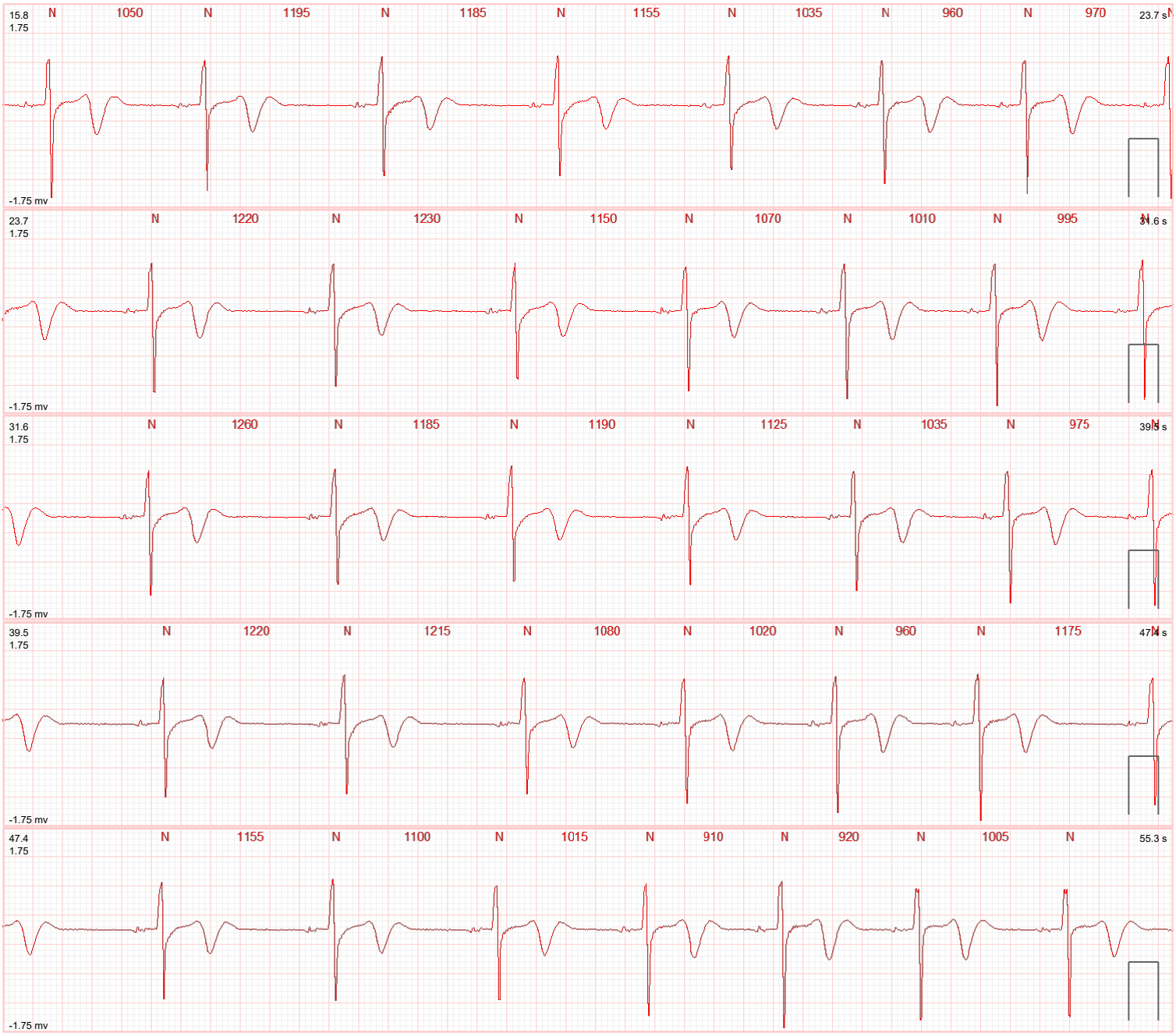
Printed: Apr 11 2025 17:03

VitalScan

ECG ANALYSIS - ECG

Cole Skotdal **Gender:** Male **Age:** 17 (DOB: Dec 5 2007) **BMI:** 23.2

Exam Date: Apr 11 2025 16:52



HPF 3.4Hz 25 mm/s - 10 mm/mV

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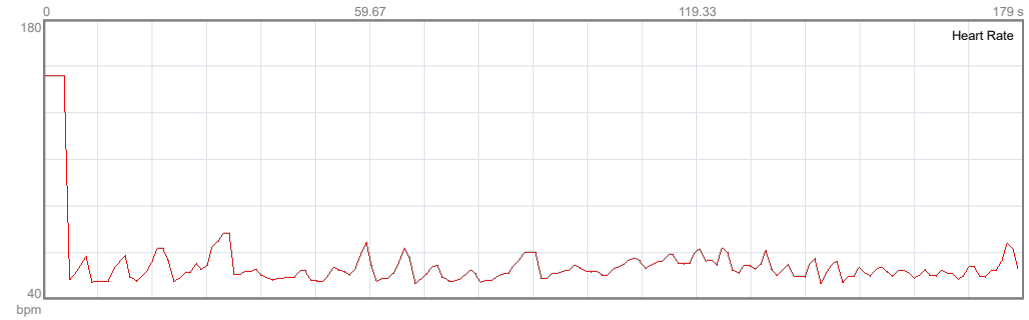
Printed: Apr 11 2025 17:03

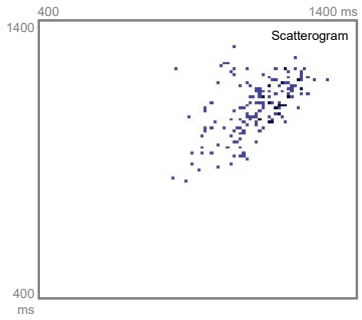
VitalScan

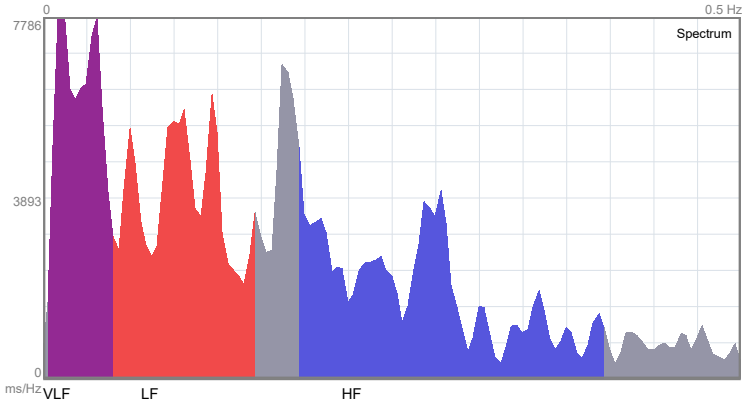
HRV CUMULATIVE STRESS ASSESSMENT

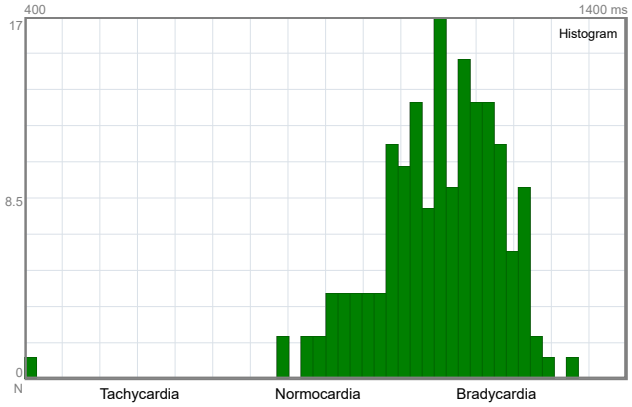
Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

Physician Only Report
Exam Date: Apr 11 2025 16:52









Total Heartbeats: 151 Artifacts: 9 (6%) Signal Quality: Results possibly compromised due to high number of artifacts!

Heart Rate: 55.87 bpm TDI: 71.27 FDI: 2984.24 SDNN: 80.47 ms

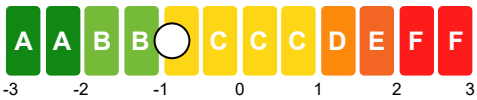
FUNCTIONAL AGE (in years): 13

HEALTH RISK Factor Based on HRV Assessment: 38.27 %

Physical Stress Coefficient: 0.34 (Normal value: 1.2 Range from 0 to 4 Unfavorable values: higher than 1.7)

PHYSICAL STRESS INDEX: -0.86

(Normal value: 0 Range from -3 to 3 Unfavorable values: higher than 1)

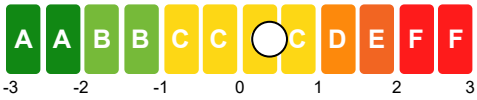


The score is below average. It indicates low physical stress. Stress in the recent past has had little or no long-term effect on your physical-health. The negative effects of physical stress are limited.

Mental Stress Coefficient: 1.42 (Normal value: 1.03 Range from 0 to 4 Unfavorable values: higher than 1.53)

MENTAL STRESS INDEX: 0.39

(Normal value: 0 Range from -3 to 3 Unfavorable values: higher than 1)



The score is average. It indicates low mental stress. Stress in the recent past has had little effect on your mental-health. The negative effects of mental stress are limited.

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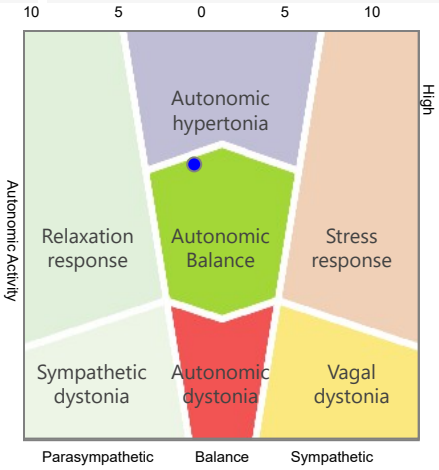
VitalScan

AUTONOMIC REGULATION ASSESSMENT

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

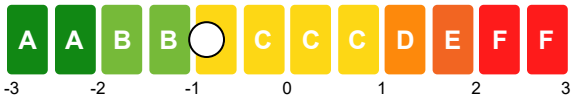
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Exam Date: Apr 11 2025 16:52





Autonomic Physical Dysregulation (PSI): -0.86

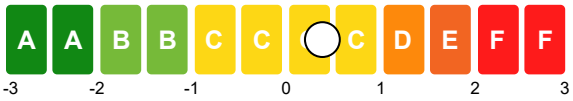
PSI is an indicator of accumulative chronic physical stress. Physical stress can be caused by long-term exposure to negative factors in one`s life or physical illness. The way in which a person copes with physical stressors is a significant contributing factor to this process. When the stress becomes too great, the body will show signs of distress.



The score is below average. It indicates low physical stress. Stress in the recent past has had little or no long-term effect on your physical-health. The negative effects of physical stress are limited.

Accumulative Mental Stress (MSI): 0.39

MSI is an indicator of accumulative mental stress. Mental stress can be caused by situations one has to cope with daily and the emotional states that result. The way in which a person copes with stressful events is a significant contributing factor to this process. When the stress becomes too great, the body will show signs of distress.



The score is average. It indicates low mental stress. Stress in the recent past has had little effect on your mental-health. The negative effects of mental stress are limited.

Abnormal Heart Beats: 9 (6%) Signal Quality: Results possibly compromised due to high number of artifacts!

Heart Rate: 55.87 bpm SDNN: 80.47 ms

QRS: 56 ms QTc: 433 ms

Ventricular Extrasystole (PVC): 0

Atrial Extrasystole (PAC): 0

FUNCTIONAL AGE (in years): 13

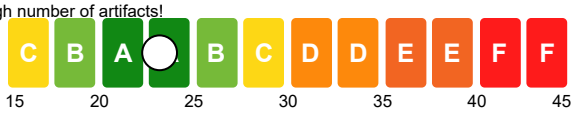
AUTONOMIC ACTIVITY: 80.44 ^

AUTONOMIC (ANS) BALANCE: 0.79 ✓

LF/HF = 4.11 ^

Borderline normal level of sympathetic and parasympathetic tone;

Body Mass Index (BMI): 23.2



Risk of Associated Disease According to BMI and Waist Size

18.5 or less: Underweight - N/A
19 - 25: Normal - very low risk of associated diseases
26 - 29: Overweight - prone to health risks
30 - 40: Overweight to Obese - high risk of associated diseases
40 or greater: Extremely Obese - very high risk of associated diseases

A = Excellent
B = Good
C = Fair
D = Poor
E = Very Poor
F = Needs Immediate Attention

Physician`s Notes:

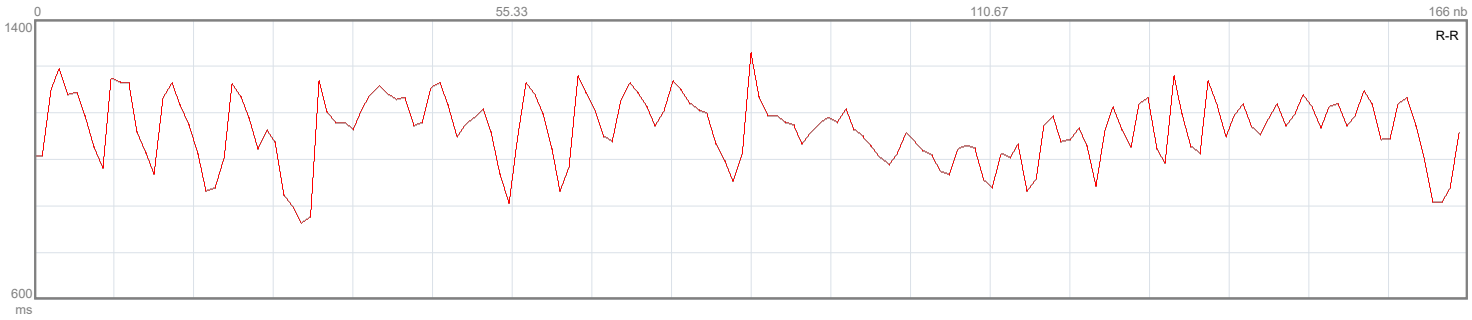
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VitalScan

HEART RATE VARIABILITY ANALYSIS

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

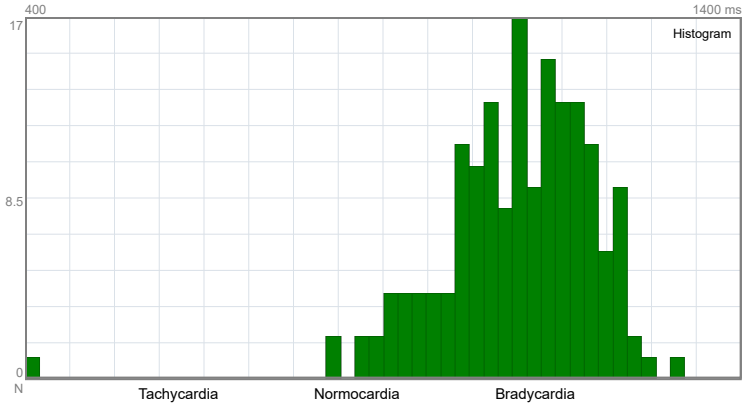
Physician Only Report
Exam Date: Apr 11 2025 16:52



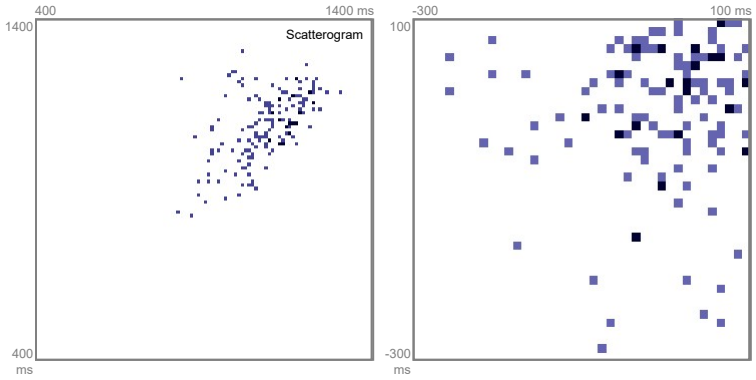
Total Heartbeats: 151 Artifacts: 9 (6%)

Signal Quality: Results possibly compromised due to high number of artifacts!

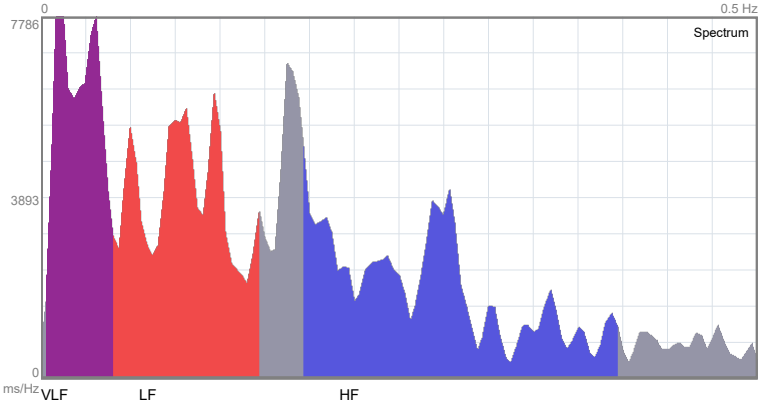
Parameters	Value	Units
Heart Rate	55.87	bpm
AMo	21	number
Mo	1080	ms
SDNN	80.47	ms
SDNN5	59.93	ms
pNN50	49.33	%
rmsSD	61.42	ms
SDSD	61.42	ms



Parameters	Value	Units
SD1	43.43	ms
SD2	105.19	ms
SD1/SD2	0.41	



Parameters	Peak(Hz)	Power(ms²)	Power(%)	Power(n.u.)
VLF	0.04	2845.04	43.03	
LF	0.1	1979.89	29.95	52.56
HF	0.2	482.07	7.29	12.8
TP		6611.71		
LF/HF		4.107		



Physician`s Notes:

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VitalScan

Arterial Stiffness Assessment - Pulse Wave Velocity

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

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Ejection Elasticity Index (EEI): 0.62

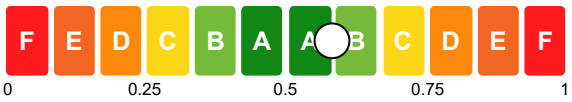
EEI is an indicator for left ventricle ejection power and elasticity of large arteries.



Borderline Normal Blood Circulation

Dicrotic Dilation Index (DDI): 0.58

DDI indicates the contractility, tension and stiffness in the small arteries.



Normal Blood Circulation

Dicrotic Elasticity Index (DEI): 0.74

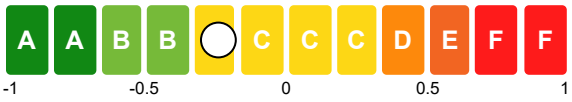
DEI represents the reflection of arterial elasticity and blood flow in the venous system.



Dilation of arterioles - Possible withdrawal of Sympathetic tone/Arterial vasodilator drugs.

Augmentation Index (AI): -0.24

Augmentation Index (AI) is a useful marker for cardiac risk. AI increases with age and a sedentary lifestyle.



AI is a measure of arterial stiffness and it provides general information about the arteries. AI is positively correlated with pulsewave velocity, blood pressure and hypertension.

A = Excellent
B = Borderline Normal
C = Mild - Moderate
D = Borderline Abnormal
E = Abnormal - Severe
F = Severe

Reflection Index (RI): 0.6



RI is an indicator of the vascular tone of the small arteries. Both vasodilation and vasoconstriction play important roles in determining vascular tone.

Stiffness Index (SI): 6.3 m/s

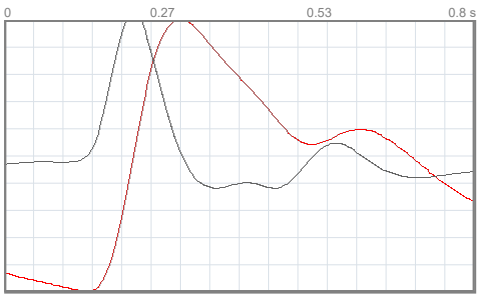


SI is a measure of large artery stiffness determined by time. SI calculation gives a value similar to aortic pulse wave velocity.

APG Pattern

An indication of the biological, (rather than chronological) age of arteries

APG Type: A



Heart Rate: 55 bpm SpO2: 97 %

C1 - Capacitive Arterial Compliance: 12.4 ml/mmHg

C2 - Oscillatory or Reflective Arterial Compliance: 6.22 ml/mmHg

Diastolic/Systolic Pressure Time Index(DPTI/SPTI): 0.41

Systemic Vascular Resistance (SVR): 793 (700-1800)

Blood Volume (BV): 6.03 l (3-5)

Cardiac Output (Q): 7.7 l/min (4.0-8.0)

Pulse Pressure (PP): 70 mmHg (25-100)

Physician`s Notes:

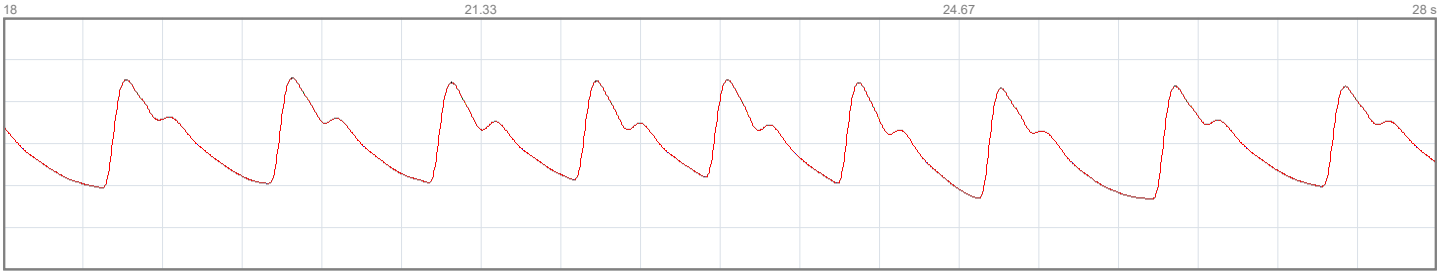
All results and analysis should be considered in the context of persons/candidate`s case history, symptoms, diagnosis, current medications, treatment plans and therapies.
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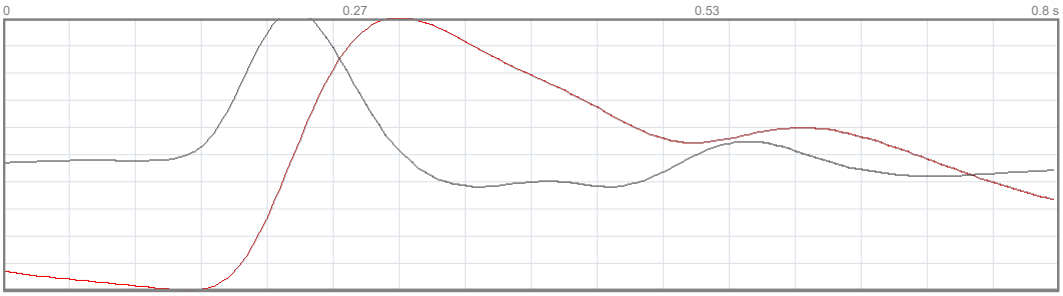
VitalScan

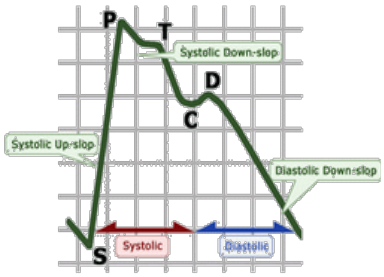
Arterial Stiffness Assessment - Pulse Wave Velocity

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

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	Score	Units		Score	Units		Score	Units		Score	Units
a-b	85	ms	**a-c**	205	ms	**a-d**	255	ms	**a-e**	330	ms
c/a	0		**e/a**	0							

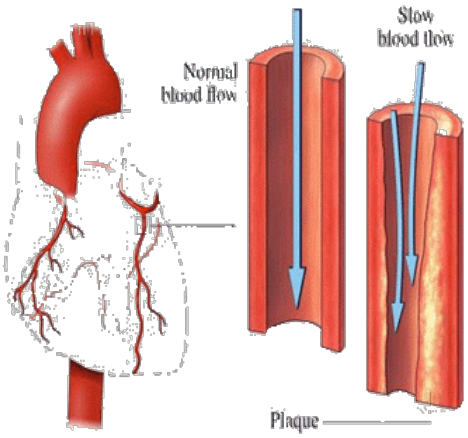
	Score	Units	Norms		Score	Units	Norms
Stroke Volume (SV)	140	ml	55-100	**Stroke Volume Index (SVI)**	64.95	ml/m2	
Cardiac Output (Q)	7.7	l/min	4.0-8.0	**Cardiac Index (CI)**	3.57	l/min/m2	2.6-4.2
Systemic Vascular Resistance (SVR)	793		700-1800	**Mean Arterial Pressure (MAP)**	77	mmHg	70-110
Pulse Pressure (PP)	70	mmHg	25-100	**Pulse Pressure Variation (PPV)**	6.3	%	
Systolic Pressure Variation (SPV)	0.3	%		**Blood Volume (BV)**	6.03	l	3-5
Pulse Transit Time (PTT)	212	ms		**Pulse Wave Velocity (PWV)**	4	m/s	2-5
Estimated PPG Cardiac Ejection Time (Etc)	385	ms	260-380	**Estimated PPG Ejection Fraction (EF)**	68	%	55-70
Pulse Height (PH)	66						

Thank you for taking the Pulse Wave Velocity (PWV) Analysis. This report gives you a quick and objective answer to how your vascular system is currently doing. The aim of these results are not to state a medical diagnosis, but to support a diagnosis by a medical professional. The result should therefore be interpreted accordingly.

PWV is an excellent analysis to evaluate vascular endothelial dysfunction. This represent the elasticity of the artery. Arteries that are atherosc lerotic, arteriosclerotic, or hardened (having reduced elasticity and increased narrowing) place an extra strain on the heart, valves, and arteries which can lead to stroke, heart attack, kidney failure and/or sudden death.

The pulse wave is a physiological phenomenon, observable and measurable in the arterial system during blood circulation. During one heart systole a certain blood volume is expelled. This propagates through the arteries due to the reciprocal transformation between kinetic energy of a segment of the expelled blood volume and the potential energy of a stretched segment of the resilient vascular wall. We can observe the changes in pressure, blood flow, velocity and profile throughout the whole pulse wave. It can be used for classification of the artery elasticity.

How is Pulse Wave Velocity measured by a finger probe?
The heart contracts and creates a direct wave which travels down the arm.
The direct wave is reflected in the lower body, and travels back towards the arm.
The direct wave and the reflected wave combine to form the finger probe.



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VitalScan

Arterial Vascular Assessment - Segmental Analysis

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

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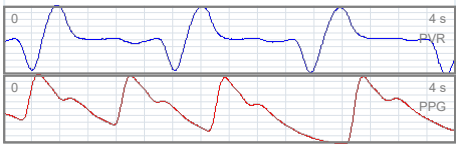
Resting - Right - Arm

EEI: 0.62 DDI: 0.59 DEI: 0.74

AI: -0.24 RI: 0.59 SI: 6.33

PTT: 210.6 ms
PWV: 4.2 m/s
SpO2: 97.1 %

DPTI/SPTI: 0.4
Systolic: 124 mmHg
Diastolic: 54 mmHg
CASP: 119 mmHg



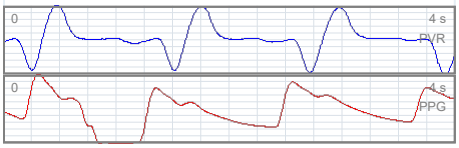
Resting - Left - Arm

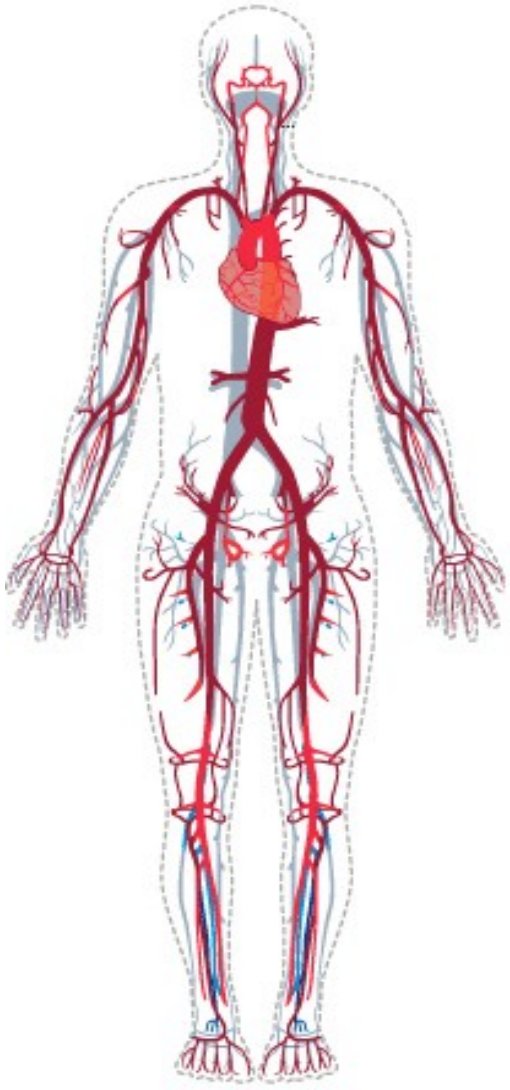
EEI: 0.63 DDI: 0.6 DEI: 0.75

AI: -0.25 RI: 0.58 SI: 6.33

PTT: 211.9 ms
PWV: 4.2 m/s
SpO2: 96.9 %

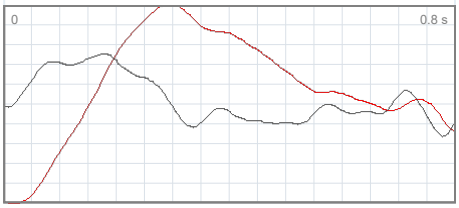
DPTI/SPTI: 0.4
Systolic: 127 mmHg*
Diastolic: 56 mmHg*
CASP: 121 mmHg*





Resting - Right - Ankle

Systolic:
Diastolic:



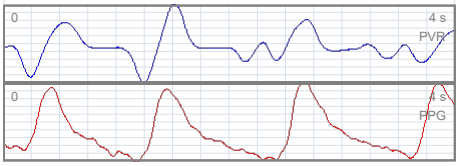
Resting - Left - Ankle

EEI: 0.56 DDI: 0.65 DEI: 0.26

AI: -0.16 RI: 0.34 SI: 7.42

SpO2: 79 %

DPTI/SPTI: 0.24
Systolic:
Diastolic:



Resting - Right - Compare

Δ Systolic:
Δ Diastolic:

Ankle/Brachial Index (ABI):

ABI < 0.4: Severe arterial disease, obstruction
0.4 - 0.7: Moderate arterial disease, obstruction
0.7 - 0.9: Some arterial disease, mild obstruction
0.9 - 1.0: Acceptable, Borderline
1.0 - 1.4: Normal range
ABI > 1.5: Abnormal Vessel hardening from PVD

TBI < 0.66: Abnormal range
0.66 - 0.75: Acceptable, Borderline
TBI > 0.75: Normal range

Resting - Left - Compare

Δ EEI: -0.07 Δ DDI: 0.06 Δ DEI: -0.48

Δ AI: 0.1 Δ RI: -0.24 Δ SI: 1.1

Δ SpO2: -17.9 %

Δ DPTI/dSPTI: -0.16
Δ Systolic:
Δ Diastolic:

Ankle/Brachial Index (ABI):

* Estimated

Physician`s Notes:

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VitalScan

Arterial Vascular Assessment - Resting - Right

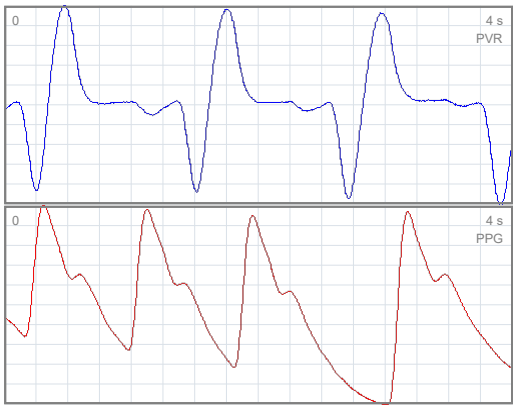
Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

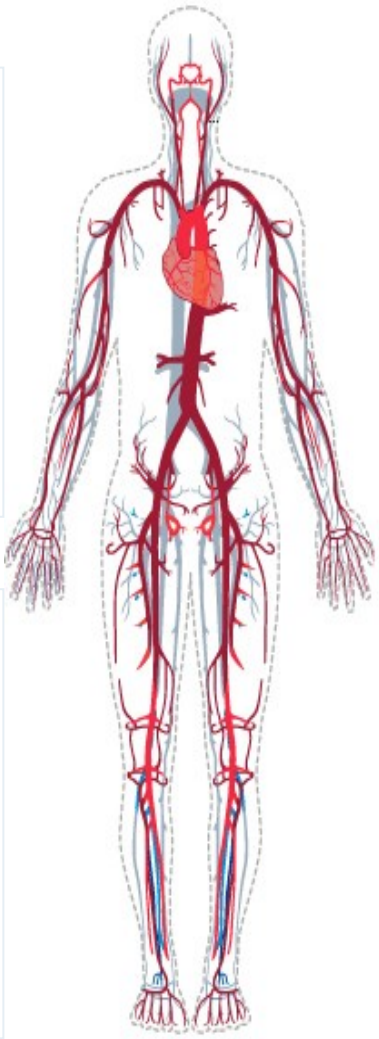
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RESTING - RIGHT - ARM

Ejection Elasticity Index (EEI): 0.62
Dicrotic Dilation Index (DDI): 0.59
Dicrotic Elasticity Index (DEI): 0.74

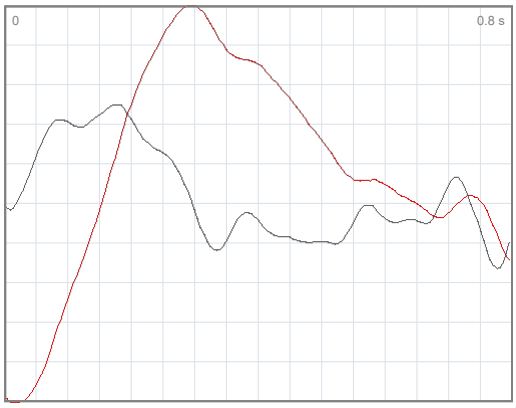
AI: -0.24	DPTI/SPTI: 0.4
RI: 0.59	Systolic: 124 mmHg
SI: 6.33	Diastolic: 54 mmHg
PTT: 210.6 ms	CASP: 119 mmHg
PWV: 4.2 m/s	SpO2: 97.1 %





RESTING - RIGHT - ANKLE

Systolic:
Diastolic:



RESTING - RIGHT - COMPARE

Ankle/Brachial Index (ABI):

Δ Systolic:
Δ Diastolic:

ABI < 0.4: Severe arterial disease, severe obstruction
0.4 - 0.7: Moderate arterial disease, moderate obstruction
0.7 - 0.9: Some arterial disease, mild obstruction
0.9 - 1.0: Acceptable, Borderline
1.0 - 1.4: Normal range
ABI > 1.5: Abnormal Vessel hardening from PVD

TBI < 0.66: Abnormal range
0.66 - 0.75: Acceptable, Borderline
TBI > 0.75: Normal range

* Estimated

Physician`s Notes:

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VitalScan

Arterial Vascular Assessment - Tilt Test

Cole Skotdal
Gender: Male
Age: 17 (DOB: Dec 5 2007)
Weight: 191 lbs
BMI: 23.2
Height: 6 ft 4 in

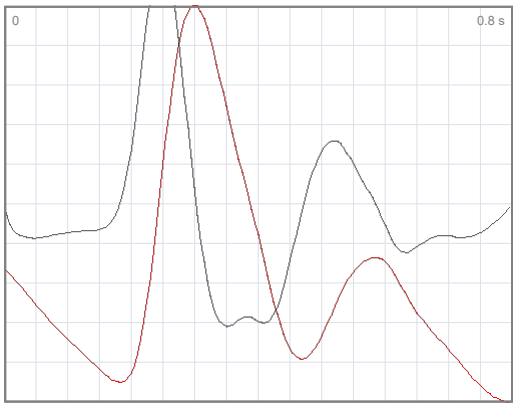
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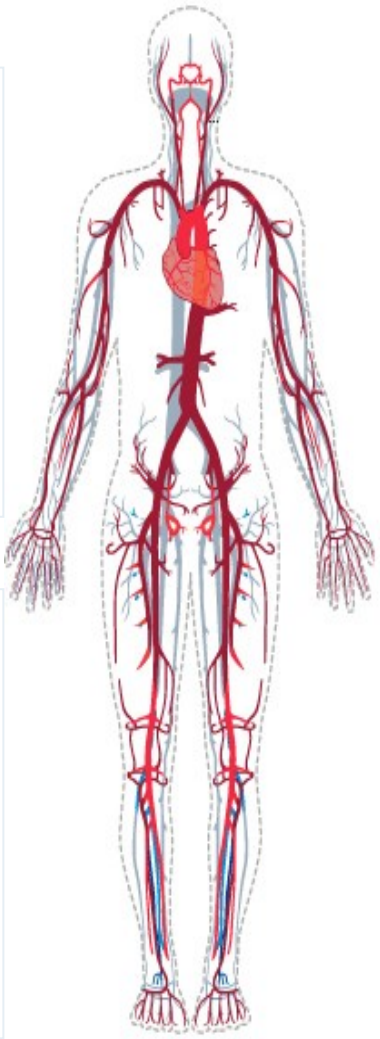
TILT TEST - ARM

Ejection Elasticity Index (EEI): 0.85
Dicrotic Dilation Index (DDI): 0.91
Dicrotic Elasticity Index (DEI): 0

AI: -0.56
RI: 0.41
SI: 7.42
PTT: 251.3 ms
PWV: 3.5 m/s

DPTI/SPTI: 0.24
Systolic: 118 mmHg
Diastolic: 59 mmHg
CASP: 113 mmHg
SpO2: 98.5 %

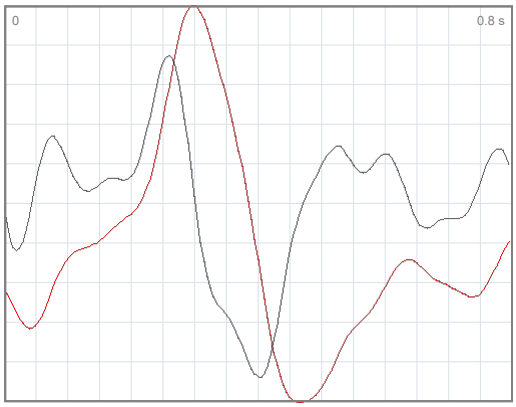




TILT TEST - ANKLE

PTT: 225 ms
PWV: 4.8 m/s

Systolic: 112 mmHg*
Diastolic: 81 mmHg*
SpO2: 80.4 %



TILT TEST - COMPARE

Ankle/Brachial Index (ABI): 0.98

Toe/Brachial Index (TBI): 1.17

ftPTT: 190 ms
ftPWV: 2.6 m/s

Δ Systolic: -3 mmHg
Δ Diastolic: 22 mmHg
Δ SpO2: -18.1 %

ABI < 0.4: Severe arterial disease, severe obstruction
0.4 - 0.7: Moderate arterial disease, moderate obstruction
0.7 - 0.9: Some arterial disease, mild obstruction
0.9 - 1.0: Acceptable, Borderline
1.0 - 1.4: Normal range
ABI > 1.5: Abnormal Vessel hardening from PVD

TBI < 0.66: Abnormal range
0.66 - 0.75: Acceptable, Borderline
TBI > 0.75: Normal range

* Estimated

Physician`s Notes:

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